

Kilonovae and *R*-process Nucleosynthesis from Neutron Star Mergers

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Outline

1. Introduction

- Question: what is the astrophysical origin of *heavy* elements?
- Answer: Neutron star mergers?

2. Modelling Nuclear physics: Which elements are synthesized in *r*-process? (proportionally)

3. Modelling Astrophysics: How much *r*-process is produced? (mass)

4. Conclusions

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The origin of elements

IA										IIB										IIA										IIIA										IVA										VA										VIA										VIIA										VIIIA																																																																																																			
1 H Hydrogen 1.008 1										2 He Helium 4.0026 2										3 Li Lithium 6.94 3										4 Be Beryllium 9.012 4										5 B Boron 10.81 5										6 C Carbon 12.01 6										7 N Nitrogen 14.007 7										8 O Oxygen 15.999 8										9 F Fluorine 18.998 9										10 Ne Neon 20.180 10																																																																																									
11 Na Sodium 22.98976928 11										12 Mg Magnesium 24.305 12										13 Al Aluminum 26.982 13										14 Si Silicon 28.085 14										15 P Phosphorus 30.974 15										16 S Sulfur 32.06 16										17 Cl Chlorine 35.45 17										18 Ar Argon 39.948 18																																																																																																													
19 K Potassium 39.0983 19										20 Ca Calcium 40.078 20										21 Sc Scandium 44.955908 21										22 Ti Titanium 47.867 22										23 V Vanadium 50.9415 23										24 Cr Chromium 51.9961 24										25 Mn Manganese 54.938044 25										26 Fe Iron 55.845 26										27 Co Cobalt 58.933 27										28 Ni Nickel 58.693 28										29 Cu Copper 63.546 29										30 Zn Zinc 65.38 30										31 Ga Gallium 69.723 31										32 Ge Germanium 72.630 32										33 As Arsenic 74.922 33										34 Se Selenium 78.971 34										35 Br Bromine 79.904 35										36 Kr Krypton 83.798 36									
37 Rb Rubidium 85.468 37										38 Sr Strontium 87.62 38										39 Y Yttrium 88.90584 39										40 Zr Zirconium 91.224 40										41 Nb Niobium 92.90637 41										42 Mo Molybdenum 95.95 42										43 Tc Technetium 98 43										44 Ru Ruthenium 101.07 44										45 Rh Rhodium 102.91 45										46 Pd Palladium 106.42 46										47 Ag Silver 107.87 47										48 Cd Cadmium 112.41 48										49 In Indium 114.82 49										50 Sn Tin 118.71 50										51 Sb Antimony 121.76 51										52 Te Tellurium 127.60 52										53 I Iodine 126.91 53										54 Xe Xenon 131.29 54									
55 Cs Cesium 132.90545196 55										56 Ba Barium 137.327 56										57-71 Lanthanides										72 Hf Hafnium 178.49 72										73 Ta Tantalum 180.94788 73										74 W Tungsten 183.84 74										75 Re Rhenium 186.21 75										76 Os Osmium 190.23 76										77 Ir Iridium 192.22 77										78 Pt Platinum 195.08 78										79 Au Gold 196.97 79										80 Hg Mercury 200.59 80										81 Tl Thallium 204.38 81										82 Pb Lead 207.2 82										83 Bi Bismuth 208.98 83										84 Po Polonium 209 84										85 At Astatine 210 85										86 Rn Radon 222 86									
87 Fr Francium 223 87										88 Ra Radium 226 88										89-103 Actinides										104 Rf Rutherfordium 261 104										105 Db Dubnium 262 105										106 Sg Seaborgium 266 106										107 Bh Bohrium 264 107										108 Hs Hassium 277 108										109 Mt Meitnerium 268 109										110 Ds Darmstadtium 271 110										111 Rg Roentgenium 272 111										112 Cn Copernicium 285 112										113 Nh Nihonium 284 113										114 Fl Flerovium 289 114										115 Mc Moscovium 288 115										116 Lv Livermorium 293 116										117 Ts Tennessine 294 117										118 Og Oganesson 294 118									
57 La Lanthanum 138.91 57										58 Ce Cerium 140.12 58										59 Pr Praseodymium 140.91 59										60 Nd Neodymium 144.24 60										61 Pm Promethium 145 61										62 Sm Samarium 150.36 62										63 Eu Europium 151.96 63										64 Gd Gadolinium 157.25 64										65 Tb Terbium 158.93 65										66 Dy Dysprosium 162.50 66										67 Ho Holmium 164.93 67										68 Er Erbium 167.26 68										69 Tm Thulium 168.93 69										70 Yb Ytterbium 173.05 70										71 Lu Lutetium 174.97 71																																							
89 Ac Actinium 227 89										90 Th Thorium 232.04 90										91 Pa Protactinium 231.04 91										92 U Uranium 238.03 92										93 Np Neptunium 237 93										94 Pu Plutonium 244 94										95 Am Americium 243 95										96 Cm Curium 247 96										97 Bk Berkelium 247 97										98 Cf Californium 251 98										99 Es Einsteinium 252 99										100 Fm Fermium 257 100										101 Md Mendelevium 258 101										102 No Nobelium 259 102										103 Lr Lawrencium 262 103																																							

NASA's Goddard Space Flight Center

The origin of elements

H1Hydrogen	The big bang Dying low-mass stars White dwarf supernovae Radioactive decay										He2Helium																	
Li3Lithium	Be4Beryllium	Cosmic ray collisions Dying high-mass stars Merging neutron stars Human-made										B5Boron	C6Carbon	N7Nitrogen	O8Oxygen	F9Fluorine	Ne10Neon											
Na11Sodium	Mg12Magnesium	Al13Aluminum	Si14Silicon	P15Phosphorus	S16Sulfur	Cl17Chlorine	Ar18Argon																					
K19Potassium	Ca20Calcium	Sc21Scandium	Ti22Titanium	V23Vanadium	Cr24Chromium	Mn25Manganese	Fe26Iron	Co27Cobalt	Ni28Nickel	Cu29Copper	Zn30Zinc	Ga31Gallium	Ge32Germanium	As33Arsenic	Se34Selenium	Br35Bromine	Kr36Krypton											
Rb37Rubidium	Sr38Strontium	Y39Yttrium	Zr40Zirconium	Nb41Niobium	Mo42Molybdenum	Tc43Technetium	Ru44Ruthenium	Rh45Rhodium	Pd46Palladium	Ag47Silver	Cd48Cadmium	In49Indium	Sn50Tin	Sb51Antimony	Te52Tellurium	I53Iodine	Xe54Xenon											
Cs55Cesium	Ba56Barium											Hf72Hafnium	Ta73Tantalum	W74Tungsten	Re75Rhenium	Os76Osmium	Ir77Iridium	Pt78Platinum	Au79Gold	Hg80Mercury	Tl81Thallium	Pb82Lead	Bi83Bismuth	Po84Polonium	At85Astatine	Rn86Radon		
Fr87Francium	Ra88Radium											Rf104Rutherfordium	Db105Dubnium	Sg106Seaborgium	Bh107Bohrium	Hs108Hassium	Mt109Meitnerium	Ds110Darmstadtium	Rg111Roentgenium	Cn112Copernicium	Nh113Nihonium	Fl114Flerovium	Mc115Moscovium	Lv116Livermorium	Ts117Tennessine	Og118Oganesson		
		La57Lanthanum	Ce58Cerium	Pr59Praseodymium	Nd60Neodymium	Pm61Promethium	Sm62Samarium	Eu63Europium	Gd64Gadolinium	Tb65Terbium	Dy66Dysprosium	Ho67Holmium	Er68Erbium	Tm69Thulium	Yb70Ytterbium	Lu71Lutetium												
		Ac89Actinium	Th90Thorium	Pa91Protactinium	U92Uranium	Np93Neptunium	Pu94Plutonium	Am95Americium	Cm96Curium	Bk97Berkelium	Cf98Californium	Es99Einsteinium	Fm100Fermium	Md101Mendelevium	No102Nobelium	Lr103Lawrencium												

[NASA's Goddard Space Flight Center](https://www.nasa.gov/goddard)

Density of objects in the Universe

Water



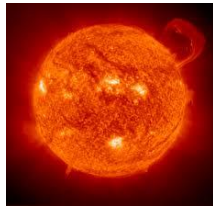
$$1 \text{ g cm}^{-3}$$

Earth



$$2 - 15 \text{ g cm}^{-3}$$

Sun



$$10 - 200 \text{ g cm}^{-3}$$

Neutron Star



$$? \text{ g cm}^{-3}$$

Density of objects in the Universe

A. $1,000 \text{ g cm}^{-3}$

B. $100,000 \text{ g cm}^{-3}$

C. $10,000,000 \text{ g cm}^{-3}$

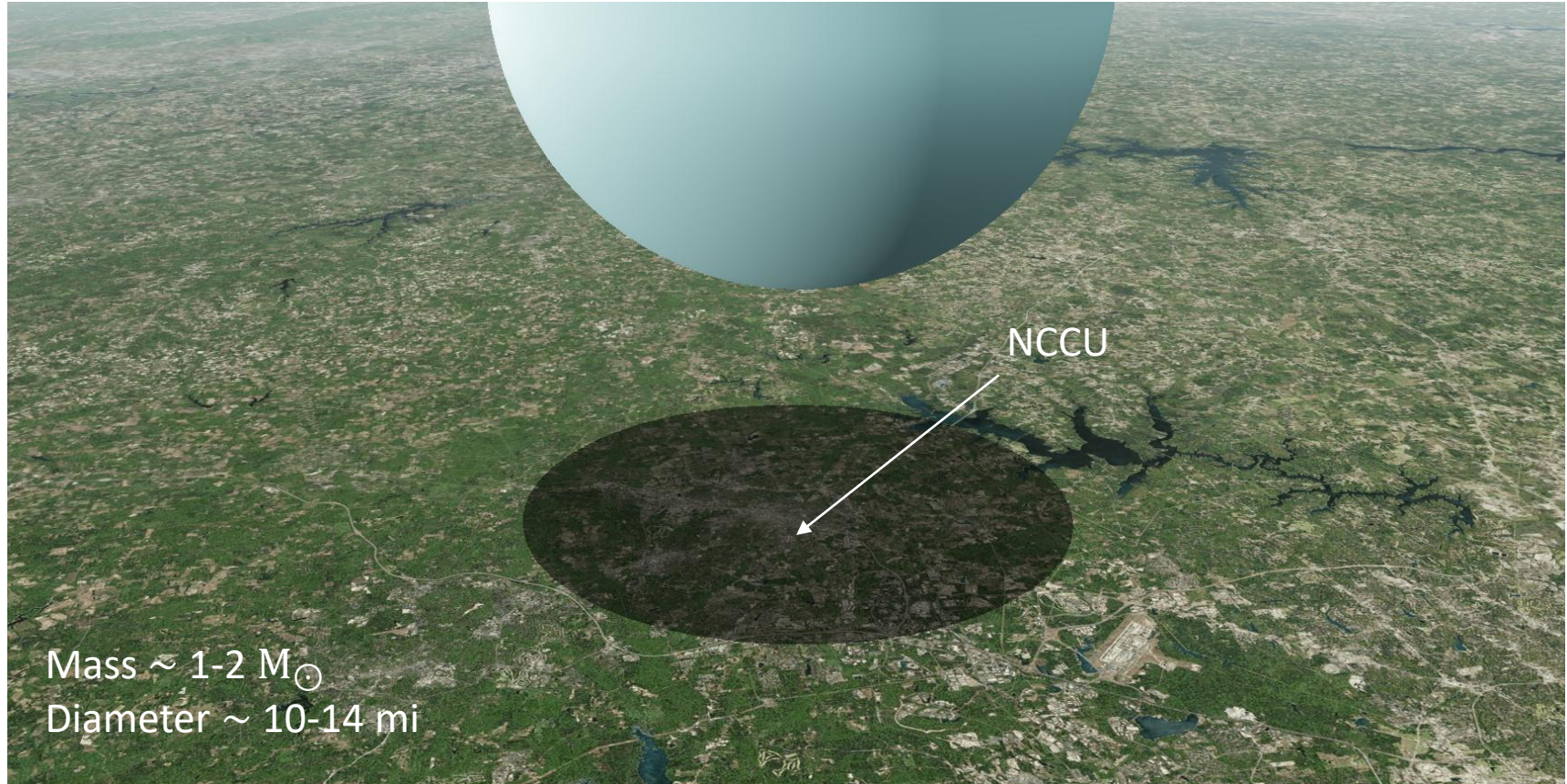
D. $1,000,000,000,000,000 \text{ g cm}^{-3} \sim 10^{15}$ ✓

Neutron Star



$10^{15} \text{ g cm}^{-3}$

Neutron star over Durham



NCCU

Mass $\sim 1-2 M_{\odot}$
Diameter $\sim 10-14$ mi

Neutron star merger



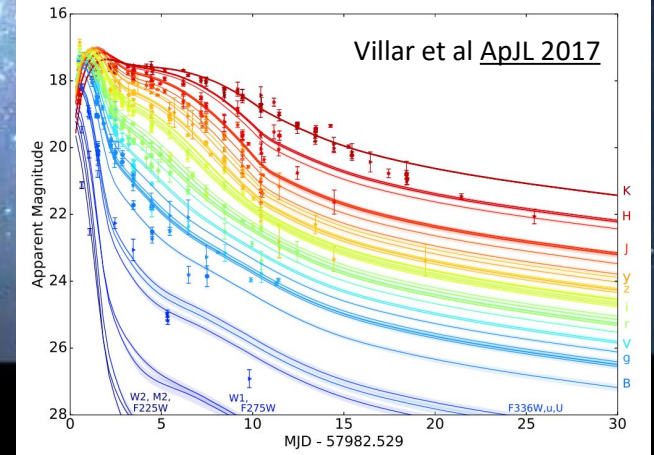
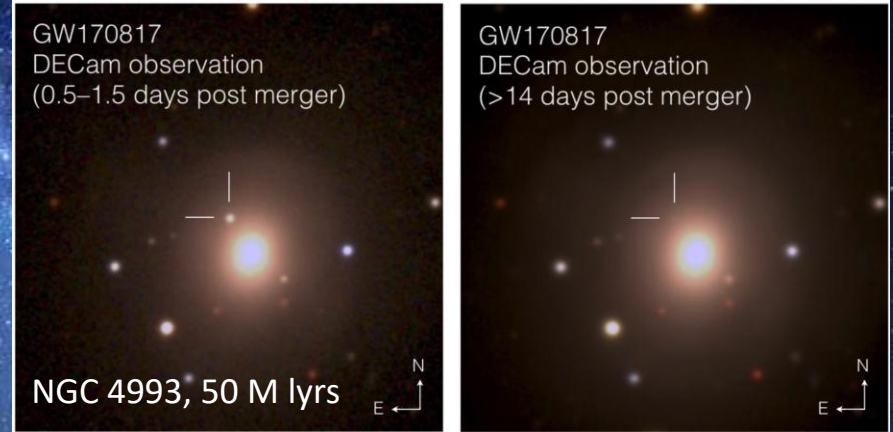
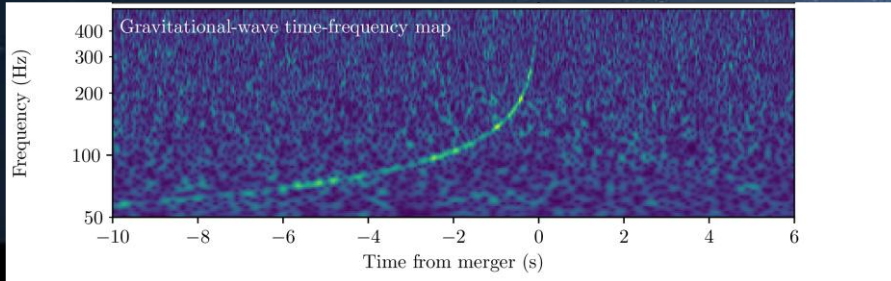
Artistic illustration from [NASA Goddard](#)

Neutron star merger: GW170817/AT2017gfo



Two perpendicular 2.5 mile arms making
a Michaelson-Morley setup

LIGO-Livingston-Louisiana, LIGO-Hanford-Washington,
Virgo-Italy, KARGA-Japan, upcoming LIGO-India (2030s)



Modelling a neutron star merger

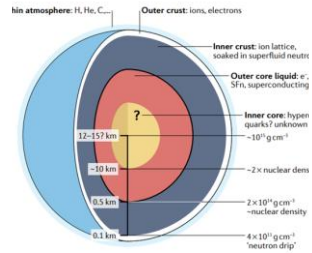
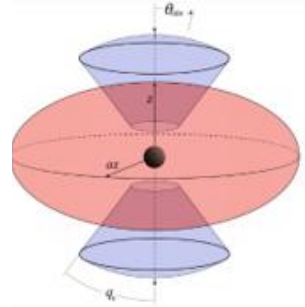


Fig. 1 | Schematic of the structure of a neutron star and its internal structure. The diagram illustrates the thin atmosphere, the outer and inner crust, and the outer and inner core, with the respective densities at different depths. Adapted with permission from the NICER Team.

Neutron star structure

Equation of state

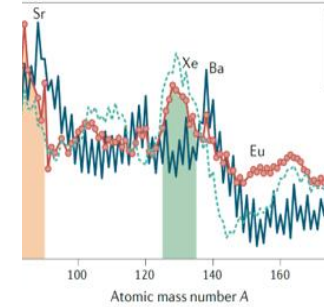
Bovard+17, Radice+18, Prakash+22,
Kedia+22, Kedia+24, Hensh+24



Hydrodynamics

General Relativistic

Breschi+21, Bulla+19,23, Heinzl+21,
Kawaguchi+20,21, **Kedia+23, Fryer+24**



Nuclear reactions

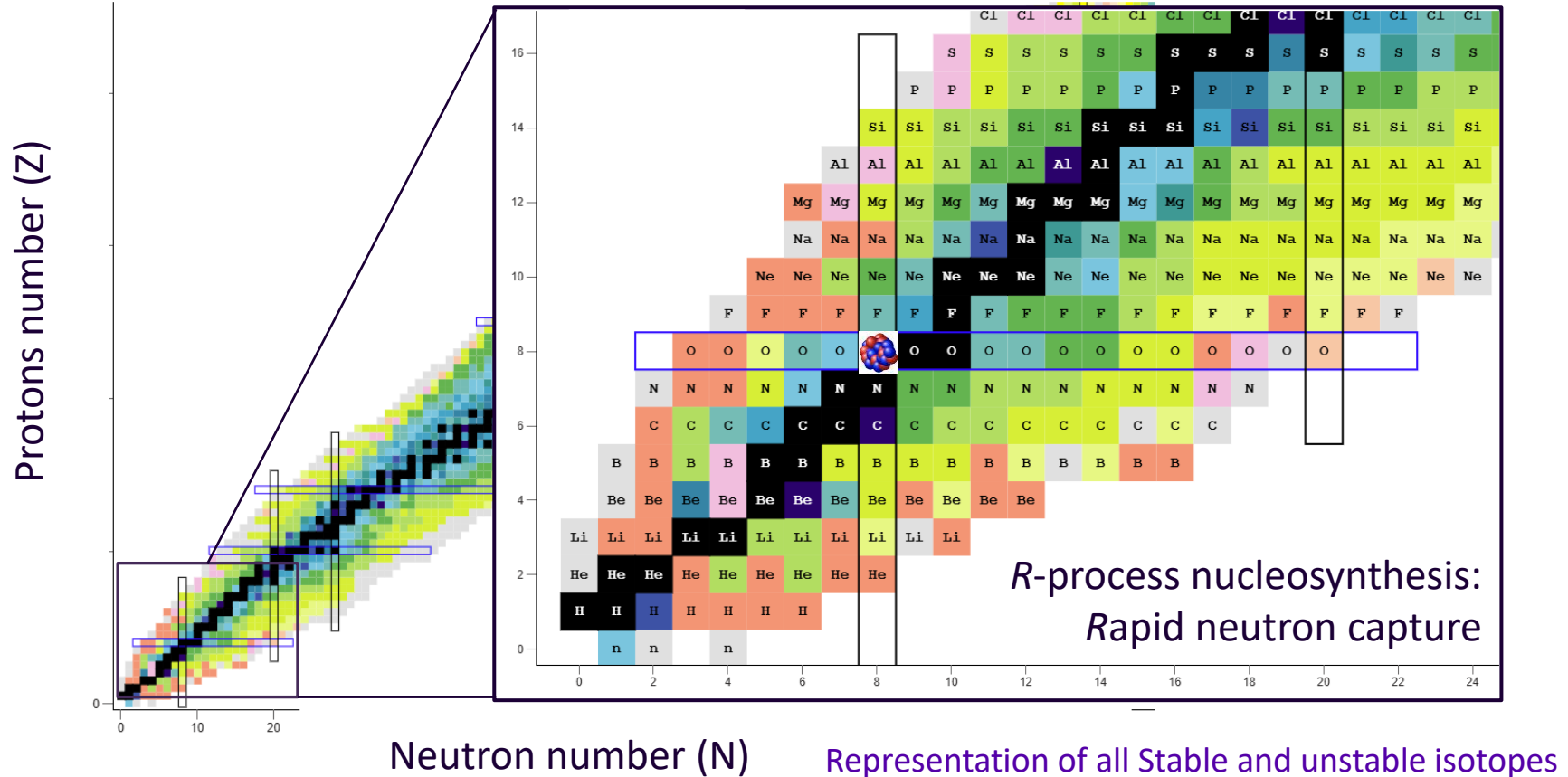
r-process nucleosynthesis

Mumpower+16, Vassh+18,+20,
Zhu+18,Holmbeck+23, **Kedia+26**

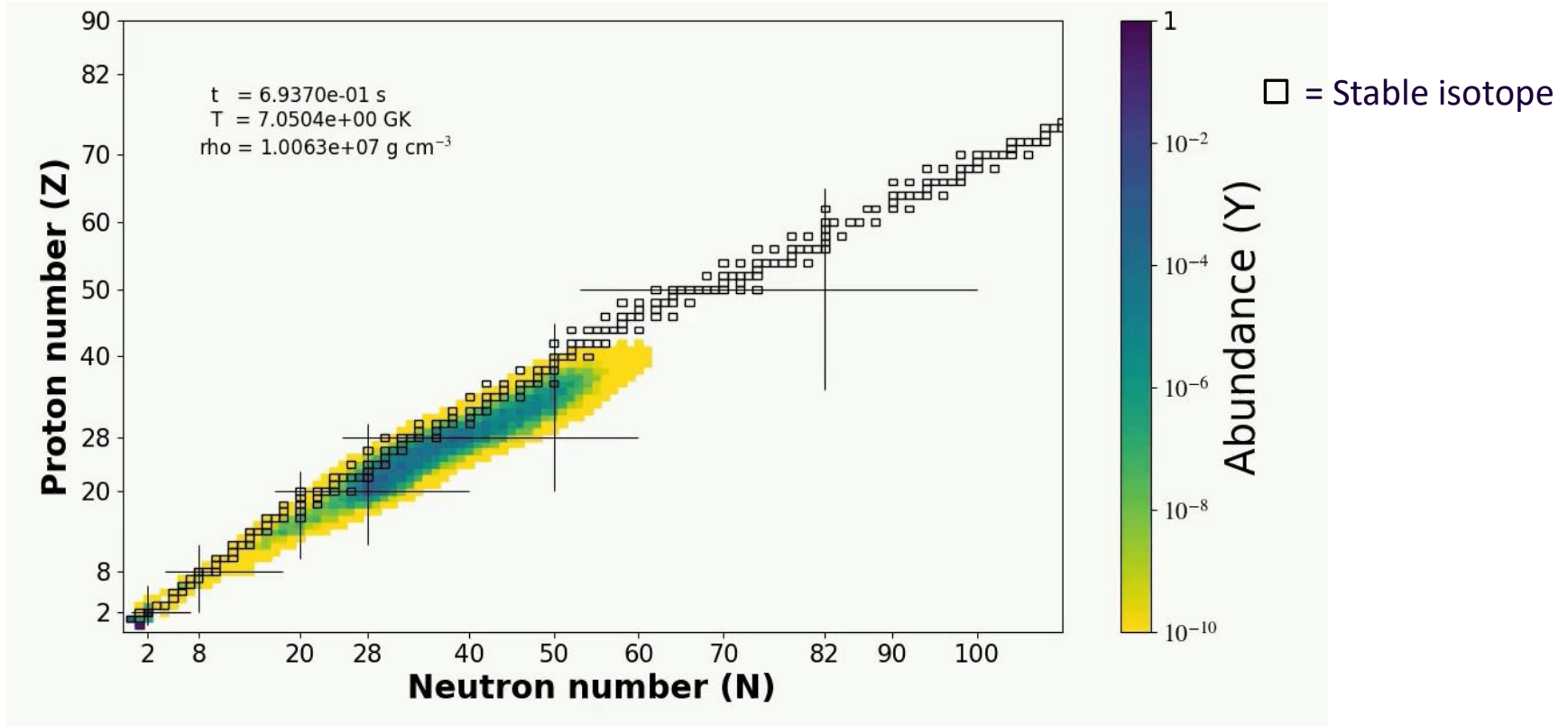
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 - Question: what is the astrophysical origin of *heavy* elements?
 - Answer: Neutron star mergers?
2. **Modelling Nuclear physics: Which elements are synthesized in *r*-process? (proportionally)**
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R-process on the Chart of Nuclides

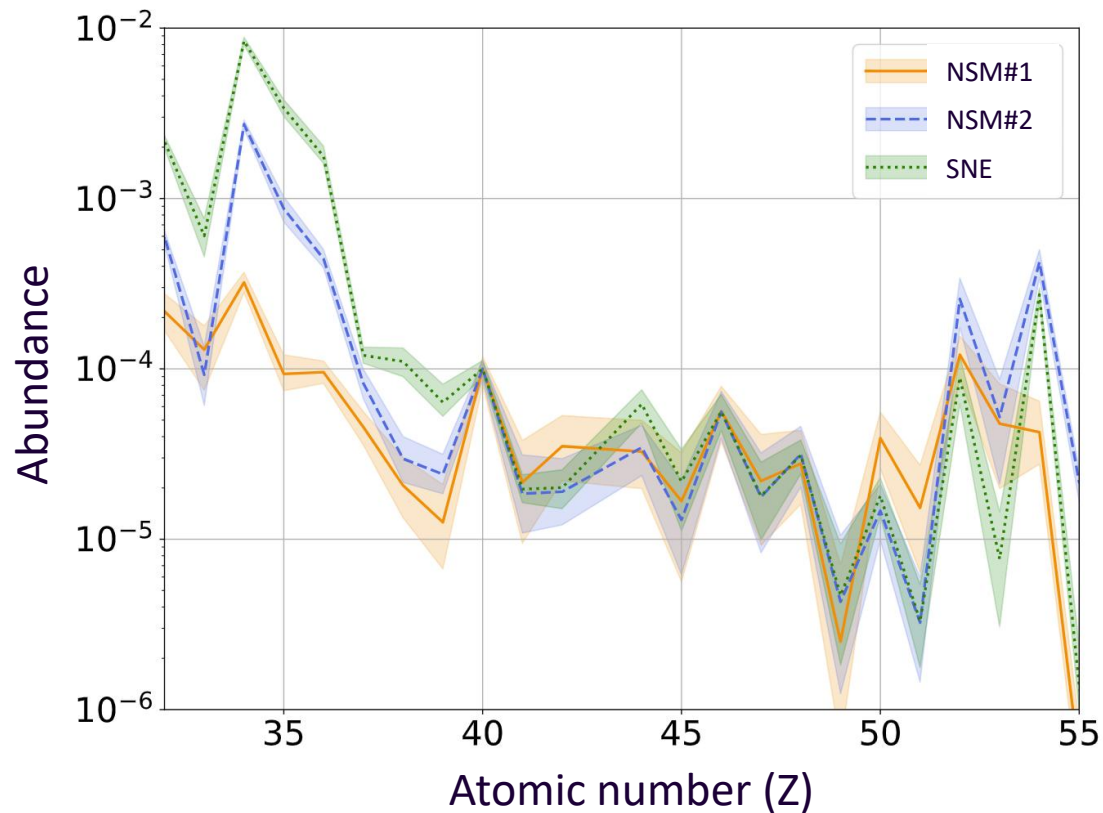


R-process evolution



Movie by AK; Calculation done on **PRISM** (reaction network code)

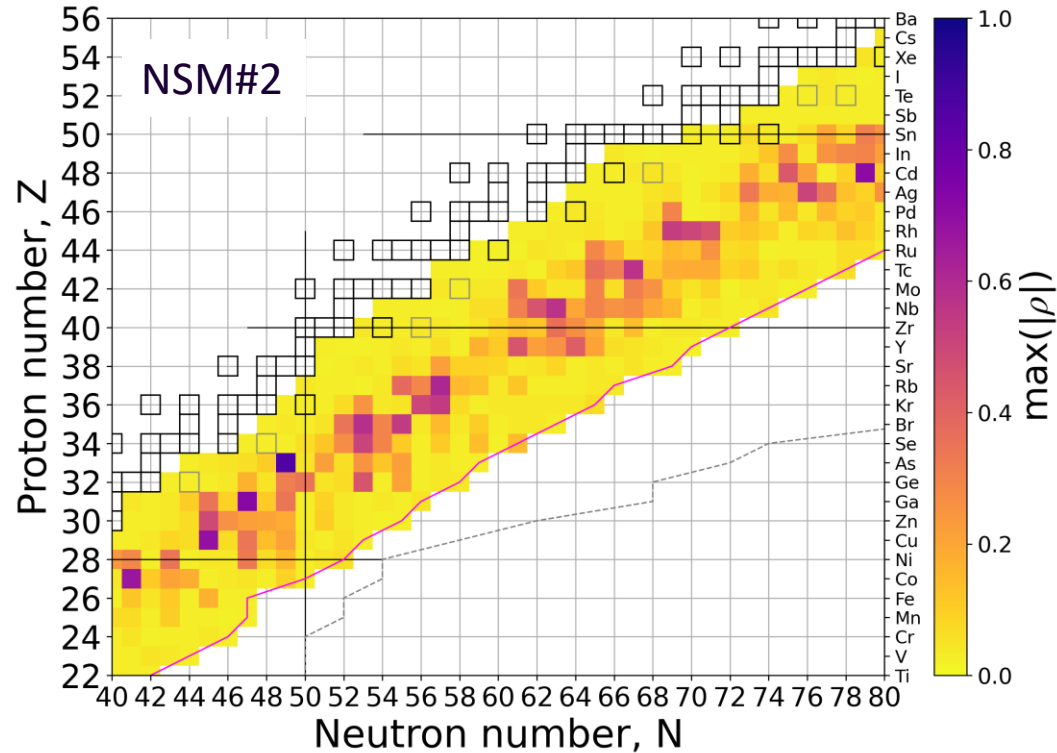
Uncertainties in r -process abundances



2σ (95 percentile) uncertainty in abundances due to uncertainties in theoretical neutron capture reaction rates.

*Shown is only the weak r -process abundance pattern.

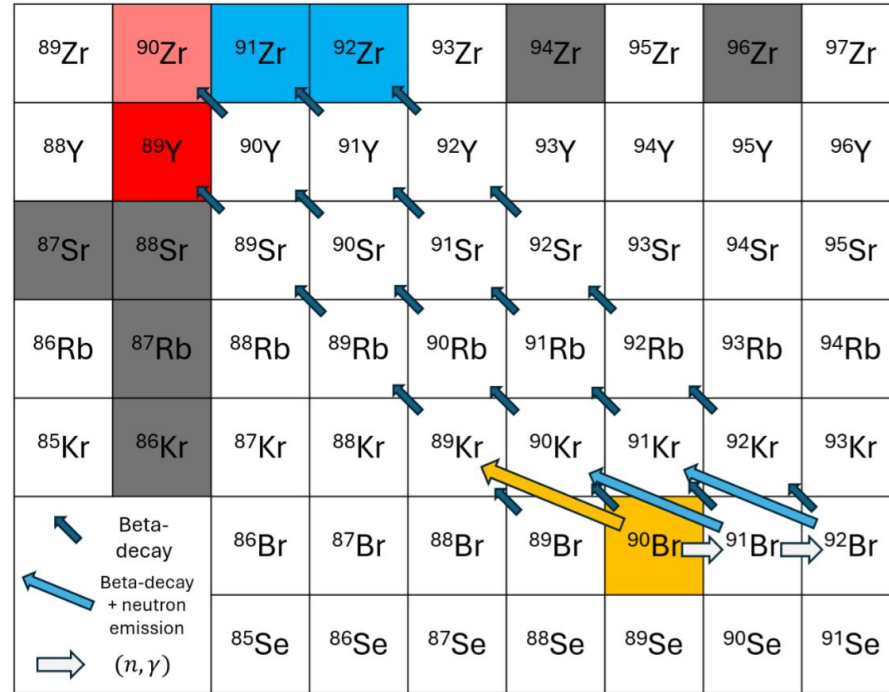
Most influential isotopes for r process



Element	(n, γ) isotope, correlation coeff. (MF14)			
Se	^{78}Ga -0.65	^{80}Ga -0.33	-	-
Br	^{78}Ga 0.72	^{80}Ga 0.34	^{81}Ge -0.32	-
Kr	^{82}As 0.86	^{82}Ge 0.35	-	-
Rb	^{87}Se -0.51	^{85}Ge -0.45	-	-
Sr	^{88}Br -0.58	^{87}Br 0.35	-	-
Y	^{90}Br -0.54	^{89}Br -0.30	-	-
Zr	^{92}Kr -0.47	^{94}Rb -0.40	-	-
Nb	^{93}Kr -0.54	^{92}Kr 0.50	^{92}Rb 0.38	-
Mo	^{94}Rb 0.61	^{100}Y -0.45	-	-
Ru	^{104}Nb -0.57	-	-	-
Rh	^{103}Nb -0.45	^{103}Y -0.42	^{103}Zr -0.36	-
Pd	^{110}Tc -0.47	^{104}Nb 0.34	-	-
Ag	^{108}Tc 0.37	^{109}Tc -0.32	^{109}Mo -0.32	-
Cd	^{110}Tc 0.57	-	-	-
In	^{114}Rh 0.53	^{115}Rh -0.50	-	-
Sn	^{116}Rh 0.46	^{124}Ag -0.34	-	-
Sb	^{123}Ag -0.51	^{123}Cd -0.44	^{120}Ag 0.33	-
Te	^{130}In -0.62	^{129}Cd 0.33	-	-
I	^{127}Cd -0.71	-	-	-
Xe	^{130}In 0.62	^{129}Cd -0.38	^{129}Sn -0.32	-

Yttrium – ^{90}Br

^{90}Br has a 25% probability of beta-delayed neutron emission and indirectly impact **Yttrium** formation



Kedia, et al., [arXiv:2602.12428 \(2026\)](https://arxiv.org/abs/2602.12428)

Reduction in uncertainties

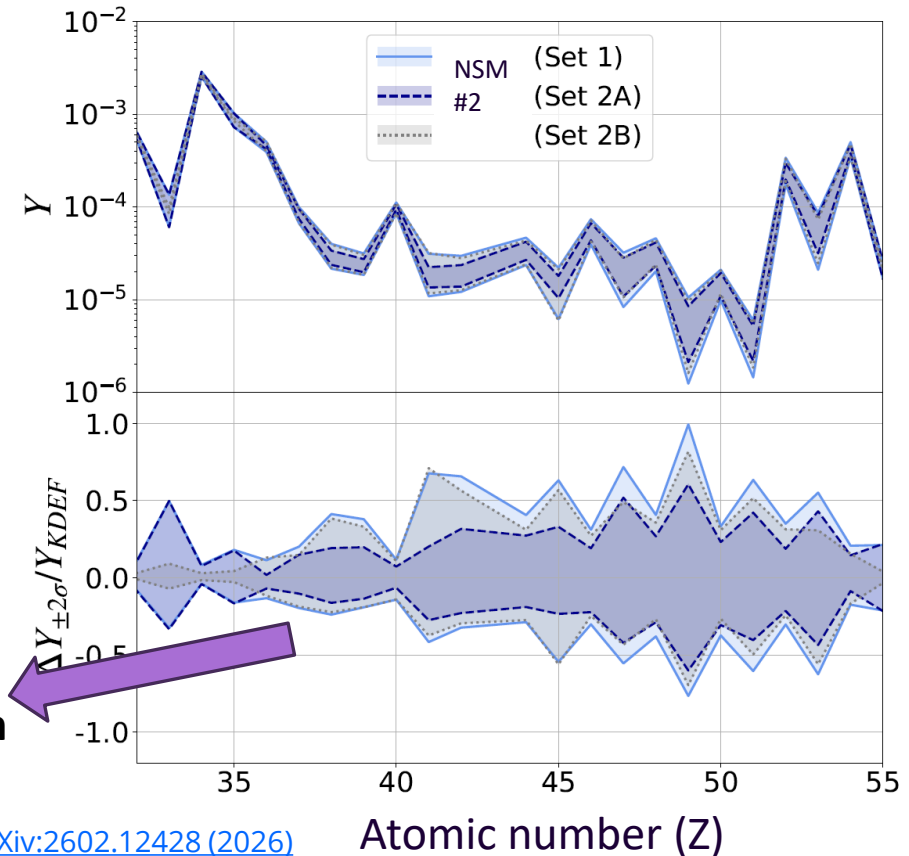
2σ (95 percentile) uncertainty bands
Hypothetical reduction in uncertainties

Set 1 : No reduction

Set 2A: Reduce uncertainties on rates for
only **50** isotopes

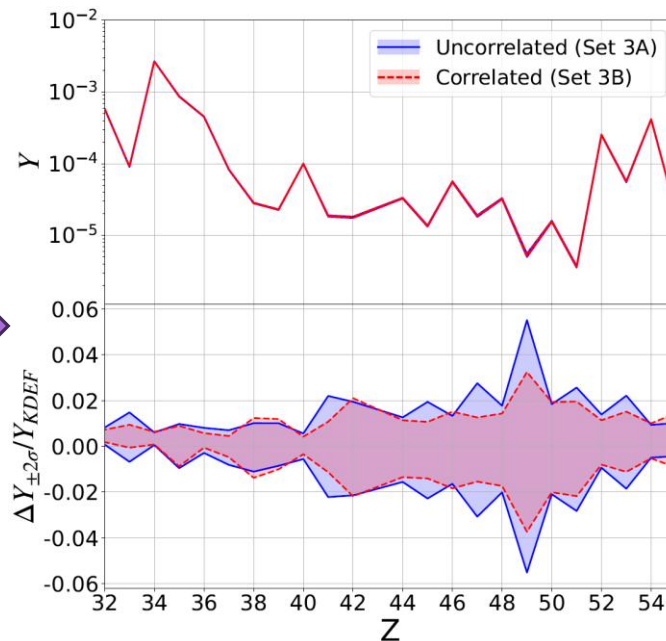
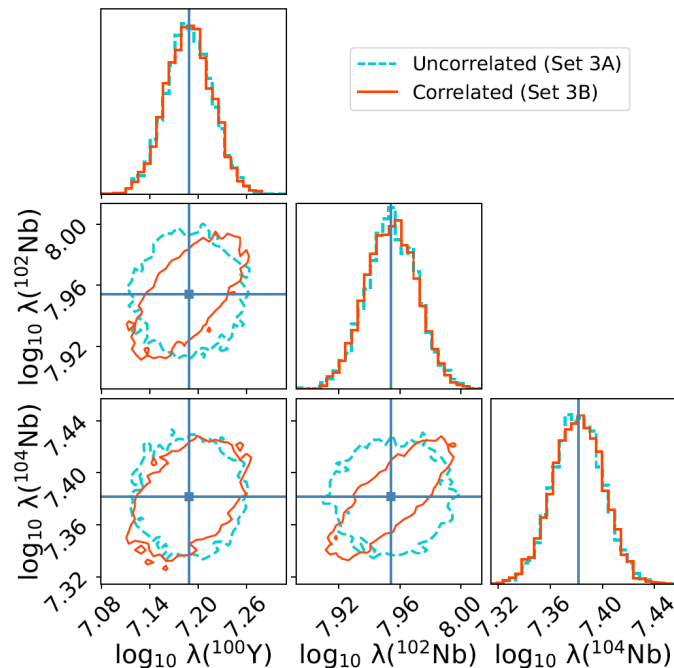
Set 2B: Reduce uncertainties on rates for
all **1000** but not for the 50
isotopes.

**The 50 isotopes lead to more reduction in
uncertainty than the remaining 1000 do!!**



Kedia, et al., [arXiv:2602.12428 \(2026\)](https://arxiv.org/abs/2602.12428)

Theoretical constraints of Reaction rates



Kedia, et al., [arXiv:2602.12428 \(2026\)](https://arxiv.org/abs/2602.12428)

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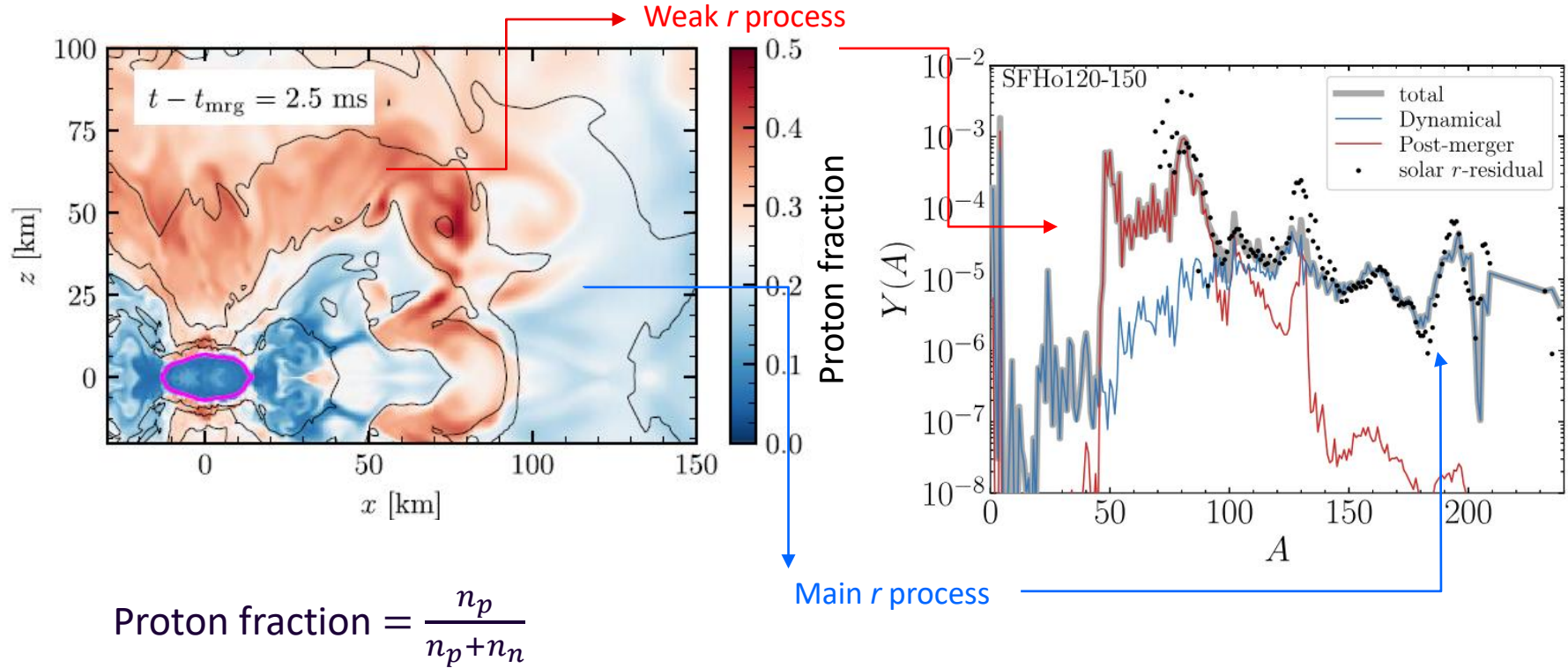
- Question: what is the astrophysical origin of *heavy* elements?
- Answer: Neutron star mergers?

2. Modelling Nuclear physics: Which elements are synthesized in *r*-process? (proportionally)

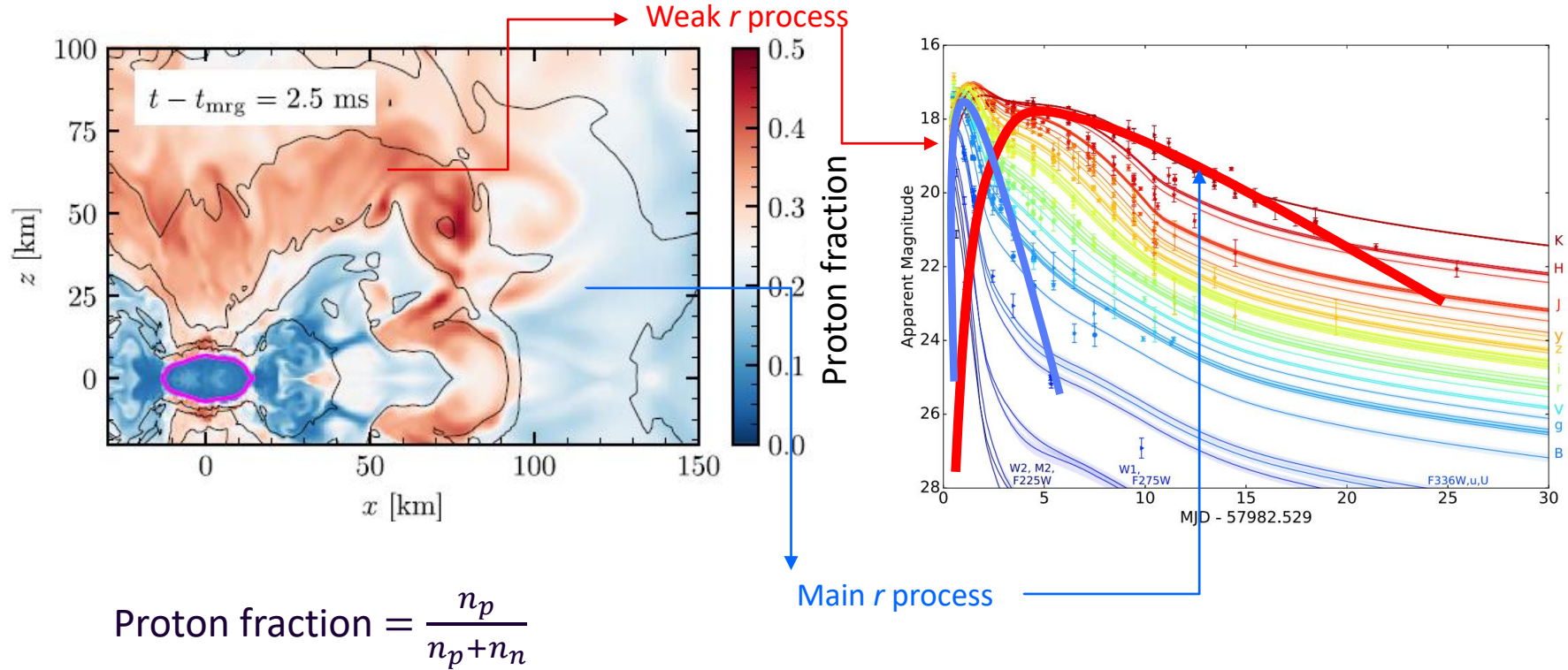
3. **Modelling Astrophysics: How much *r*-process is produced? (mass)**

4. Conclusions

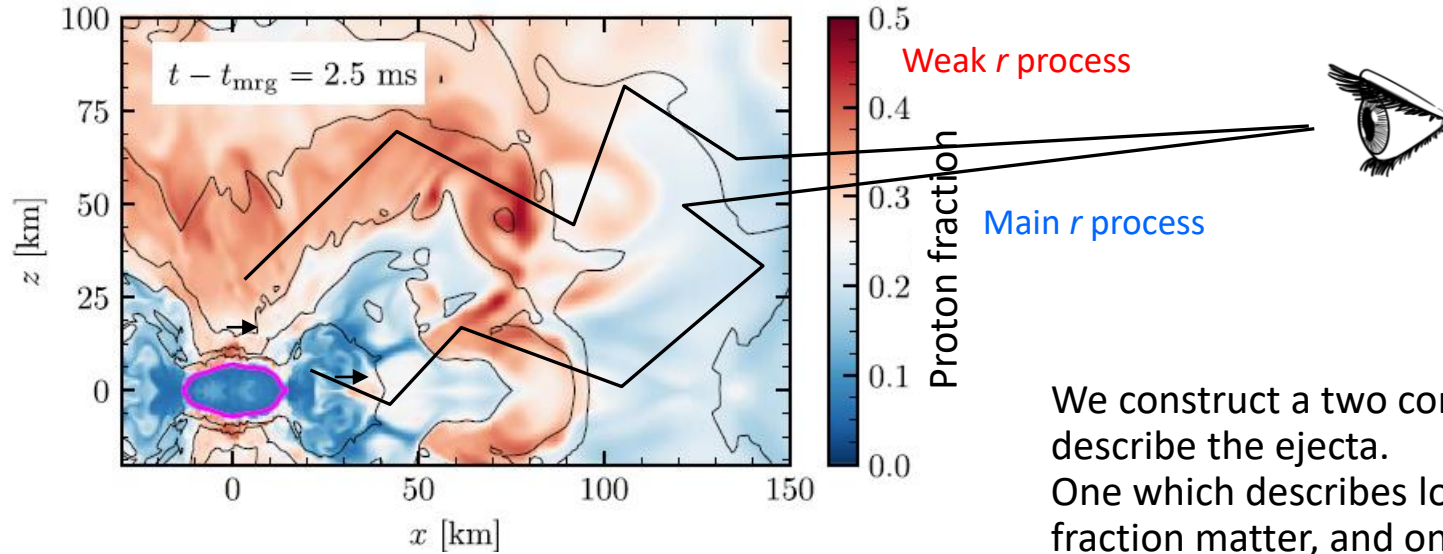
Abundance dependence on ejecta



Kilonova contribution of the ejecta



Kilonova contribution of the ejecta

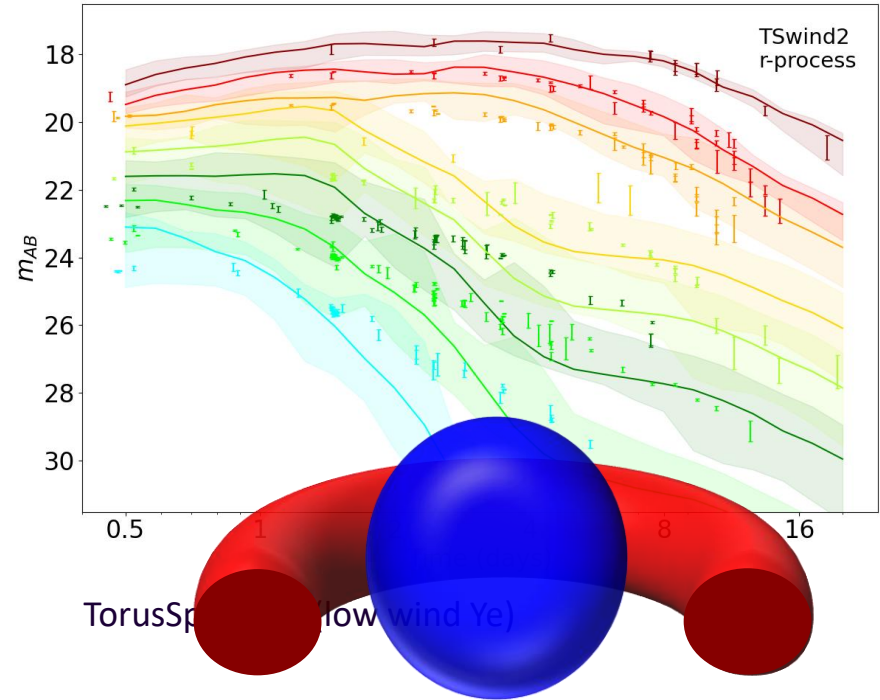
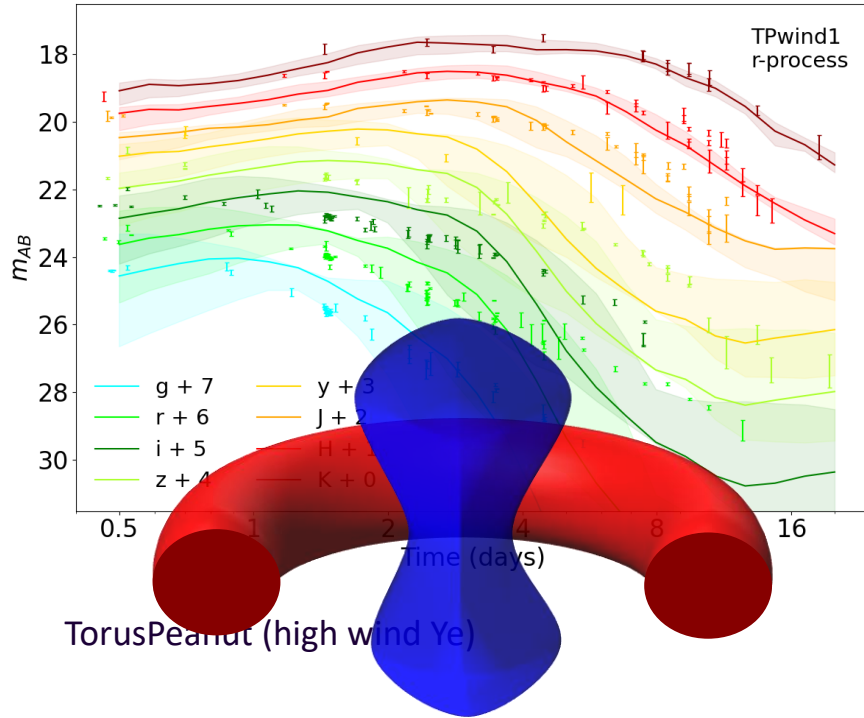


We construct a two component to describe the ejecta. One which describes low proton fraction matter, and one which has a high proton fraction matter.

Radiative transfer model

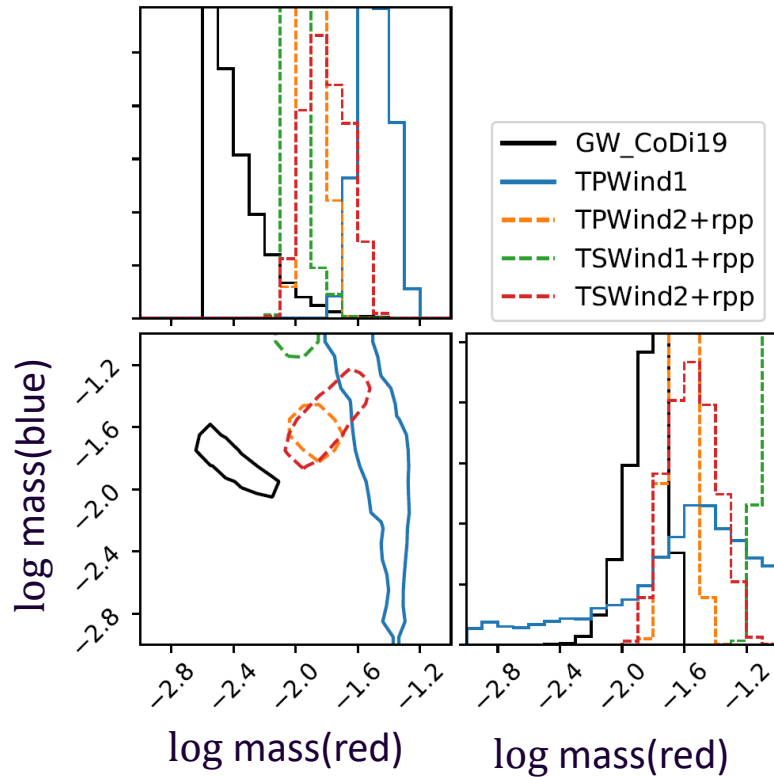
- Radiative transfer **SuperNu**. (Wollaeger et al 2013, 2014, Wollaeger, .., Kedia, Zenodo, 2023)
- Nucleosynthesis and heating from **WinNet**. (Winteler et al. 2012, Korobkin et al. 2012)
- Thermalization efficiencies (Barnes et al. (2016))
- Atomic opacities (Fontes et al. 2020)

Kilonova Light curves

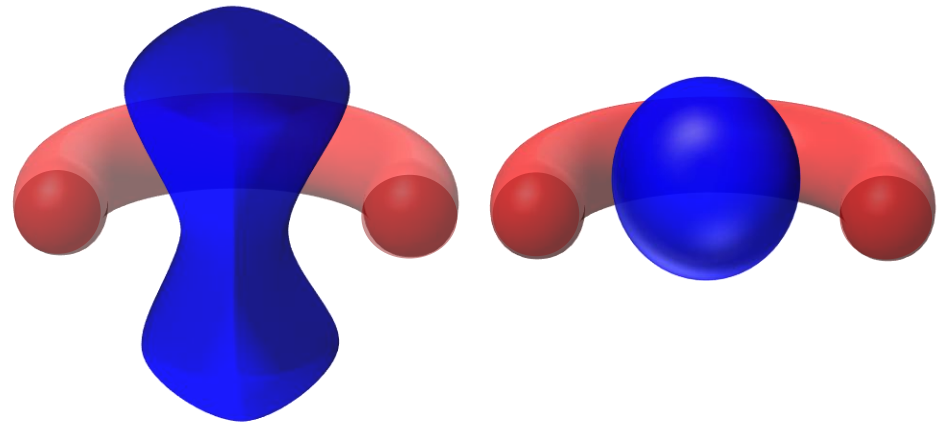


AK et al Phys. Rev. Research 5, 013168 (2023)

Bayesian inference based ejecta masses from GW170817

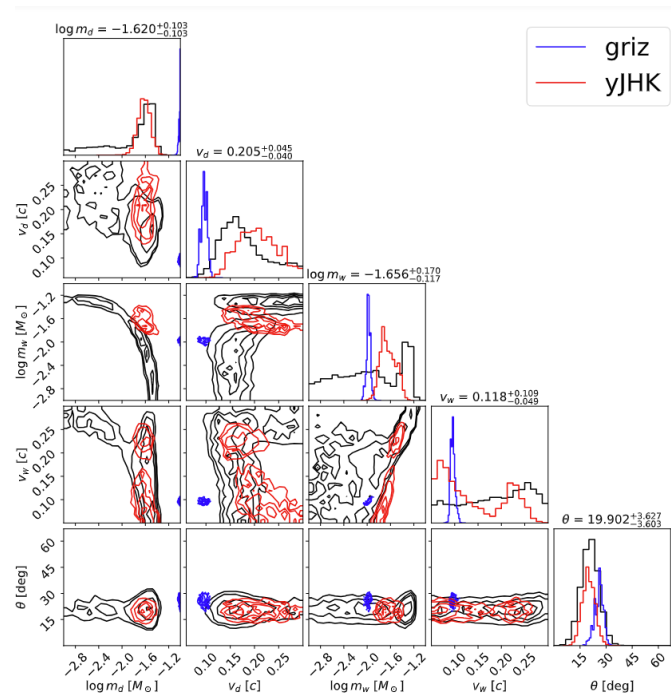
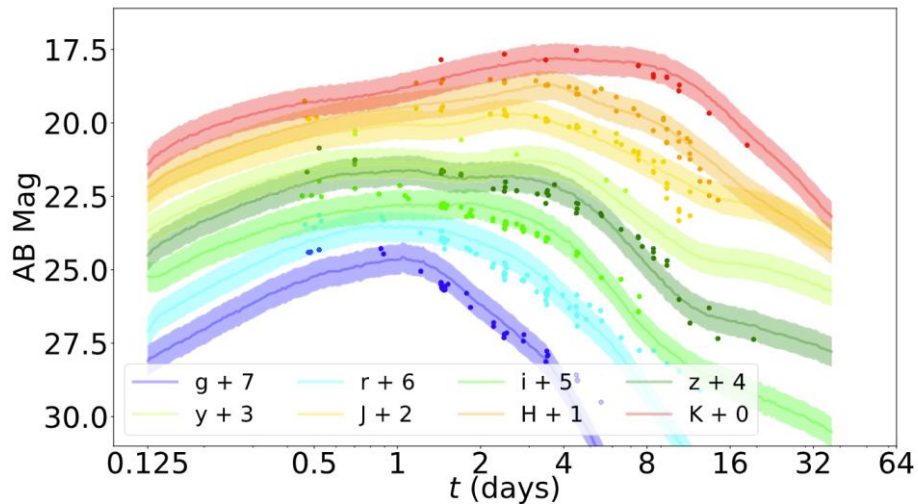


General agreement between the GW and kilonova observations on the inferred r -process enriched ejecta. However, how much mass was ejected depends on the morphology.



AK, et al, Phys. Rev. Res. 5, 013168 (2023);

Light curves w/ Neural Networks



Yinglei Peng*, Ristic, Kedia, et al. [Phys. Rev. Res. 6, 033078 \(2024\)](#)

*Conducted as undergrad at URochester, currently a graduate student at GeorgiaTech

Inference dependence on chosen wavebands.

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- Question: what is the astrophysical origin of *heavy* elements?
- **Answer: Neutron star mergers -> Maybe**

2. Modelling Nuclear physics: Which elements are synthesized in *r*-process? (proportionally)

3. Modelling Astrophysics: How much *r*-process is produced? (mass) (Answers: earlier slides)

4. Conclusions

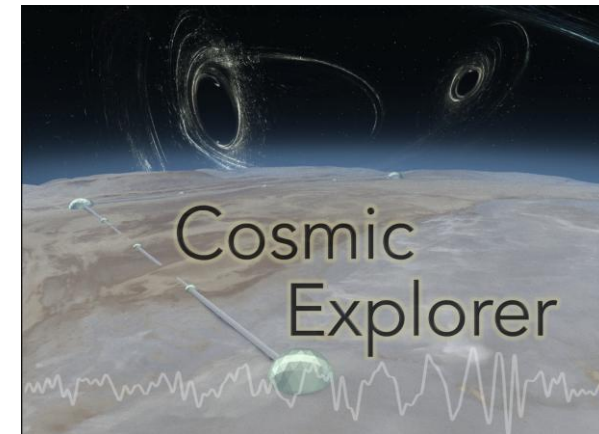
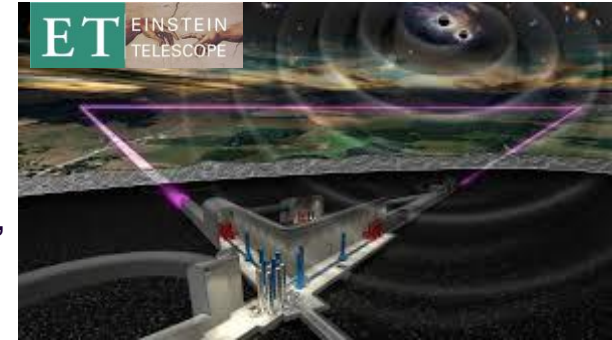
Future of observations and experiments



Nuclear experiments at:
TUNL (Triangle), FRIB (MSU),
ARIEL (TRIUMF)

EM observation:
Nancy Grace Roman Space Telescope,
Vera Rubin Observatory,
James Webb Space Telescope

GW observation:
LIGO-Virgo-KAGRA
3rd Generation GW detectors



Summary

- **Origin of elements**
- **Neutron star mergers — GW170817**
 - Supernovae
- ***r*-process nucleosynthesis**
 - Uncertainties in abundance from theoretical models
 - Isotopes that reduce uncertainties
- **Kilonova modelling**
 - Modelling in the two component (red, blue) method
 - Uncertainties in the amount of ejected mass produced.

Outlook

- ***r*-process nucleosynthesis**
 - Constraints on nuclear reaction rates
 - Astrophysical environments: Supernovae, Neutron star mergers, Collapsars, etc.
- **Kilonova modelling**
 - Apply hydrodynamic data into kilonova model
 - Model asymmetries such as due to neutron star-black hole merger.
 - Nuclear uncertainties in radioactivity

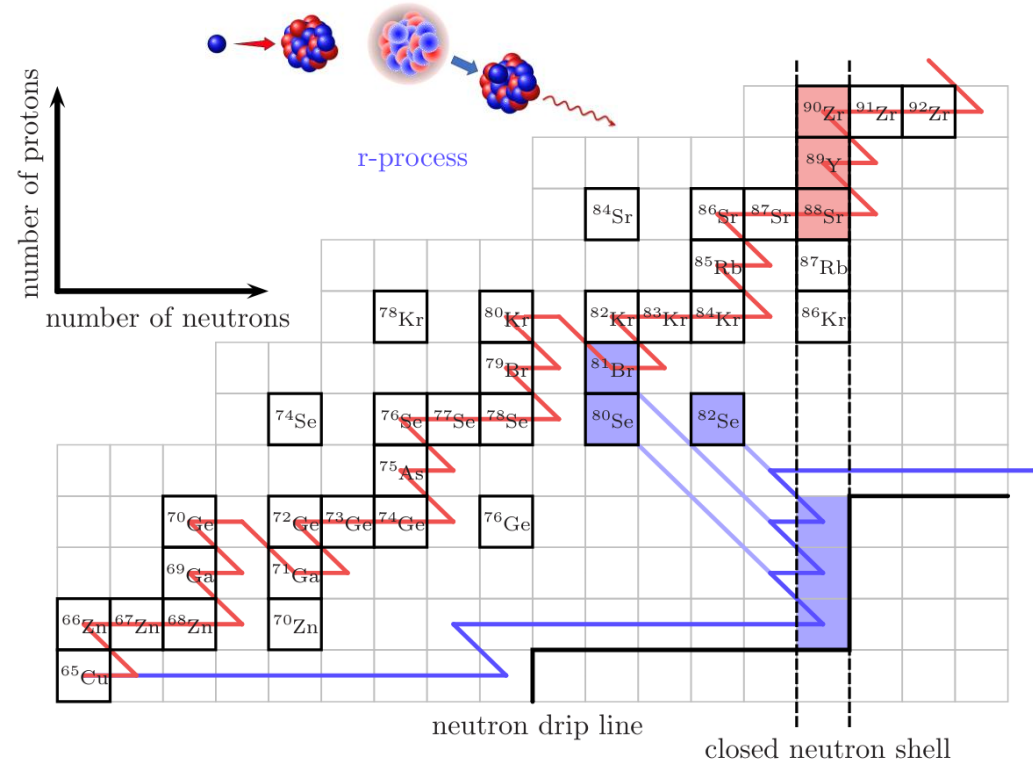
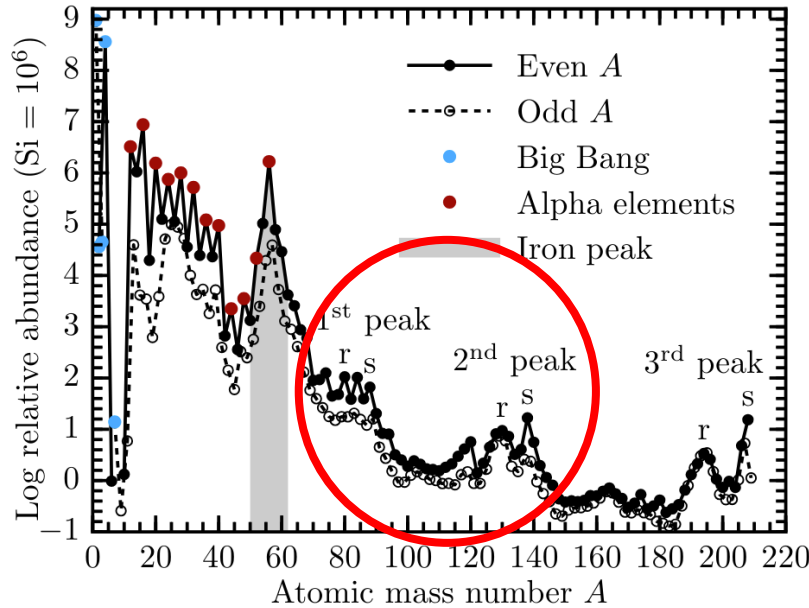
More theory, observational, and experimental work in nuclear astrophysics.



askedia@ncsu.edu

Extra slides

Solar chemical composition And the r process



Figures from [Lippuner \(2018\)](#)

r -process: $\tau_{n\text{-capture}} \ll \tau_{\beta\text{-decay}}$

Astrophysical sites for the r process

Binary Neutron star mergers:

(Meyer+1989, Metzger & Fernandez 14, Just+15, Radice+18, Kedia+22, Lund+24)

Neutron star-black hole mergers:

Surman+08, Darbha+21, Curtis+23

Magnetorotational supernovae:

Nishimura+15, Mosta+18, Reichert+21, Zha+24

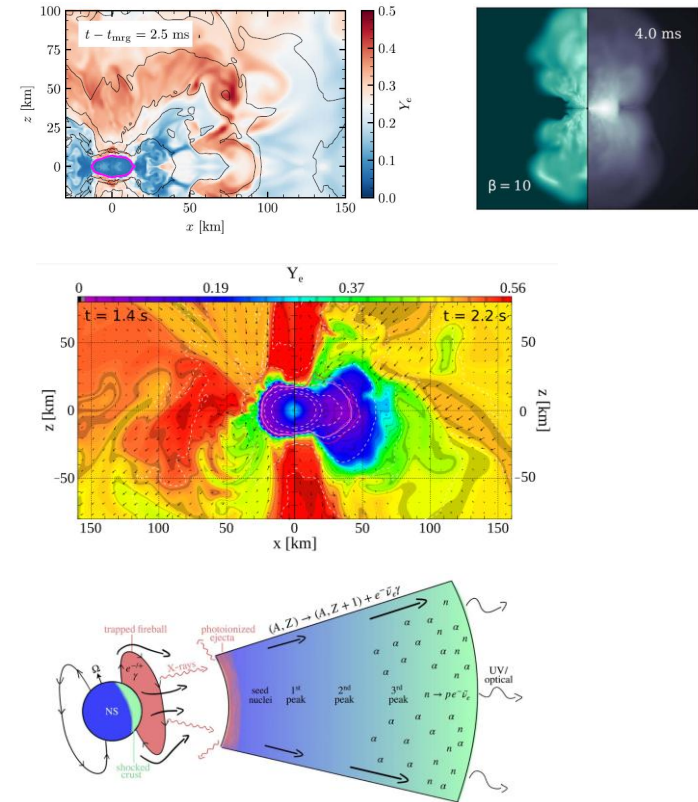
Magnetar giant flares:

Patel+25a, 25b

Collapsars:

Siegel+2019, Anand+2024

and **Others exotic scenarios**

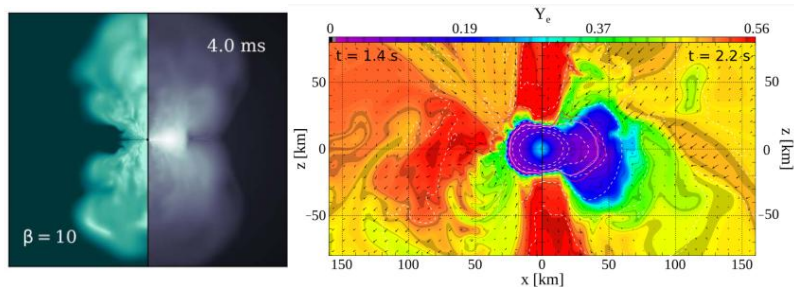


Astrophysics and nuclear physics inputs

Astrophysics:

Magnetized and non-magnetized disk winds from neutron star mergers
(K. Lund et al 2024, Metzger & Fernandez 2014)

Magnetorotational supernova
(Reichert et al 2021)

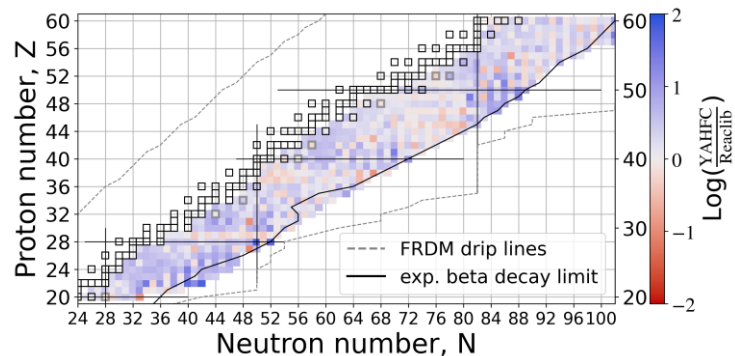


Nuclear physics:

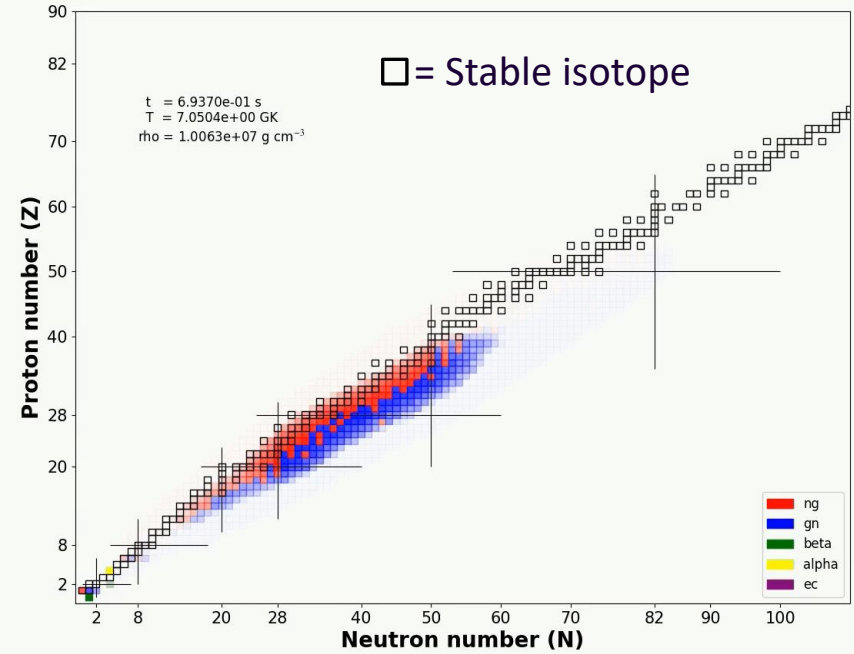
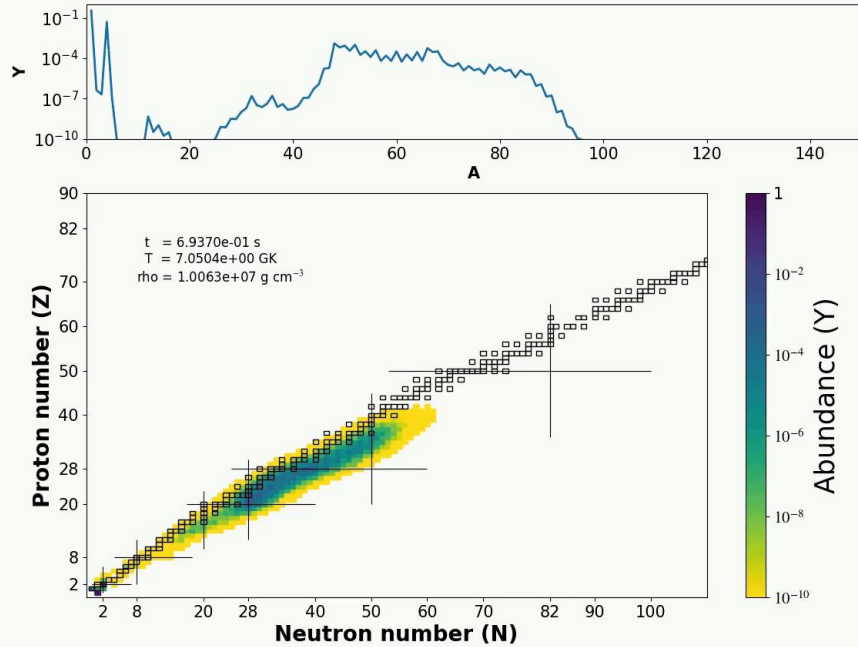
Neutron capture rates from YAHFC
(Yet Another Hauser-Feshbach Code)
(Ormand, Kravvaris, LLNL)

1000 isotopes, $Z=20-60$;

Beta-decay rates



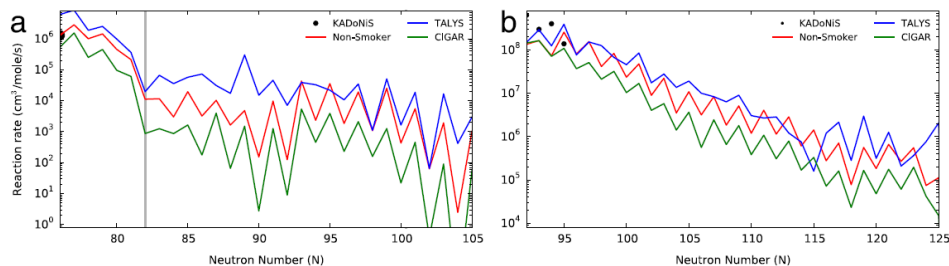
R-process evolution



Movies by AK; Calculation done on **PRISM** (reaction network code)

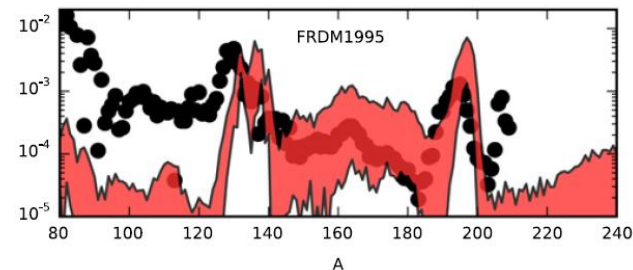
ng -> neutron capture
gn -> photodissociation (neutron removal)
beta -> beta decay

Uncertainties in neutron capture reactions



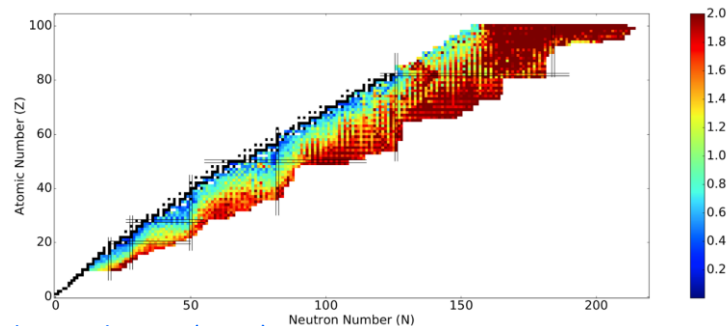
Sn –chain

Eu –chain



[Mumpower+, PNP \(2016\)](#) ;

multiplicative factors: $10^{-3} - 10^3$ in n-cap rates;
Mumpower et al ApJ 970:173 (2024)

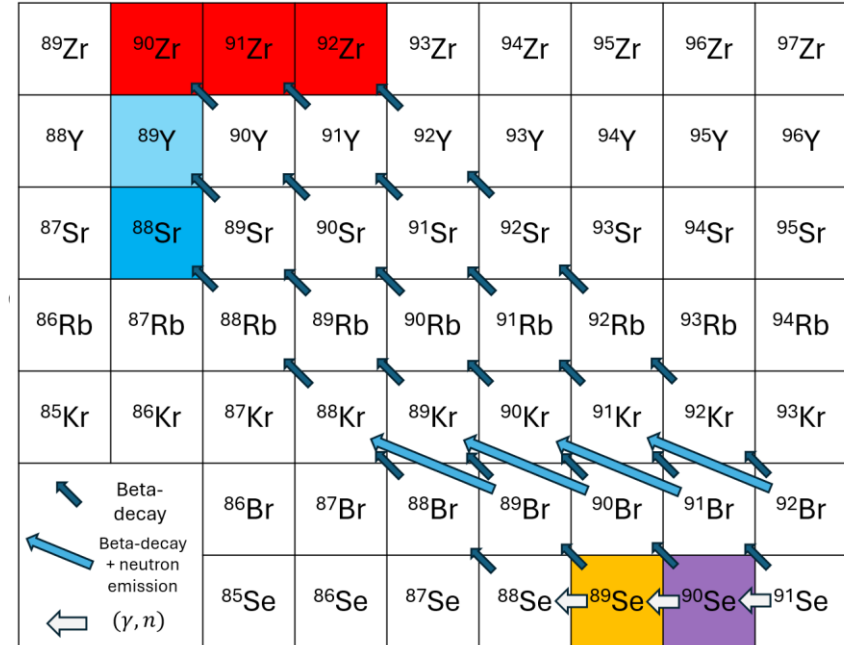
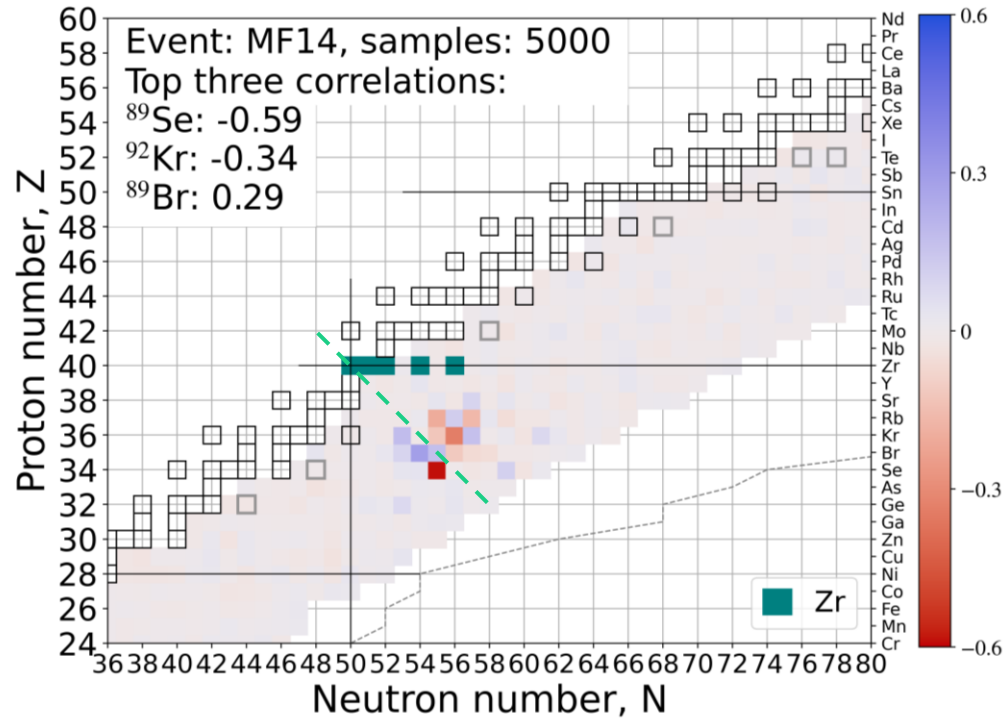


[Nikas et al, arXiv \(2020\)](#)

N-capture uncertainties

N-cap uncertainties propagated to abundances

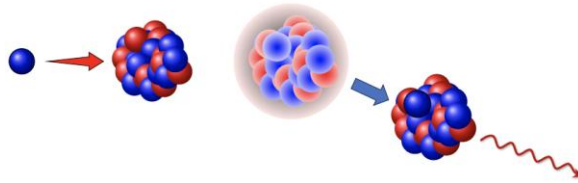
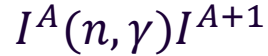
Elemental abundance — isotopic neutron capture rate correlation



Due to detailed balance: $\lambda_{\gamma, n}^{Z, A+1} \propto \lambda_{n, \gamma}^{Z, A}$

Nomenclature: Reactions at play in r -process

Neutron Capture



Photodissociation



Beta-decay



Hauser-Feshbach statistical model

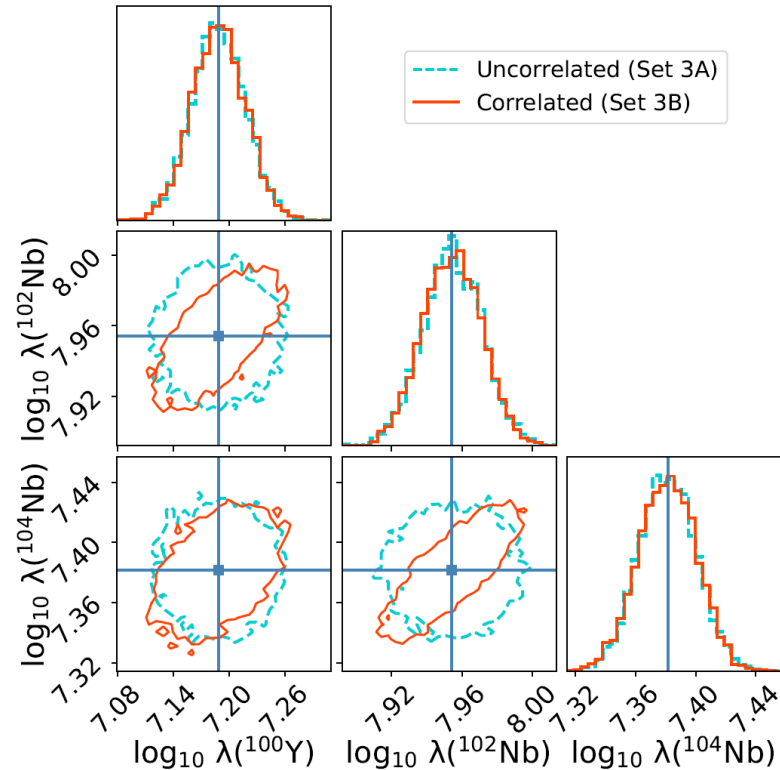
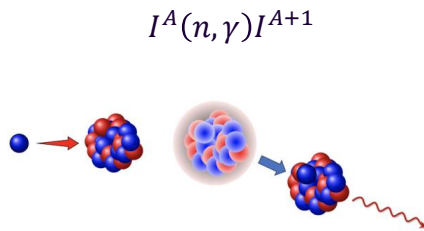
Hauser-Feshbach statistical model inputs:

☐ Mass Model

☒ Optical Model Potential

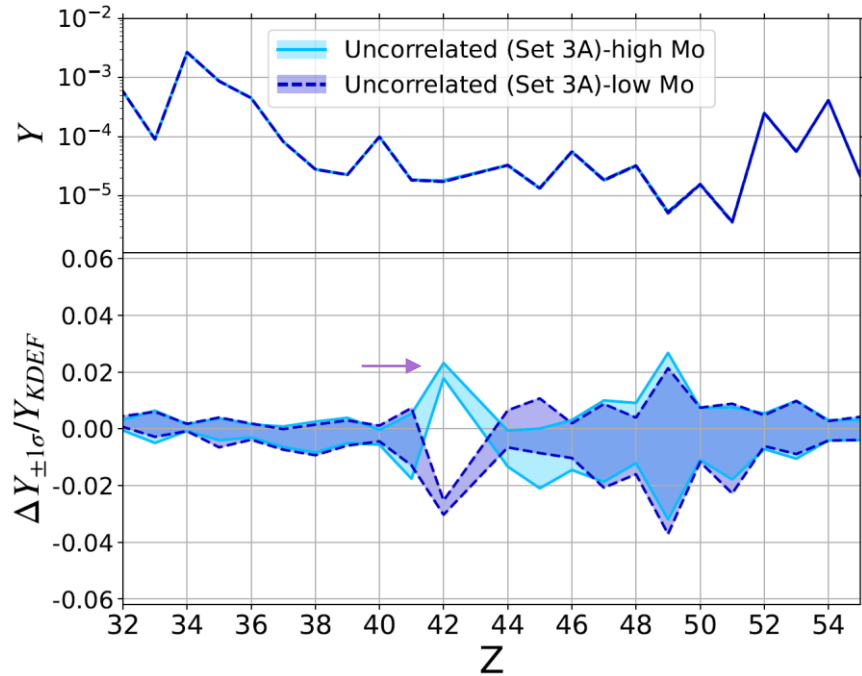
☐ Level Density

☐ Gamma-Strength Function

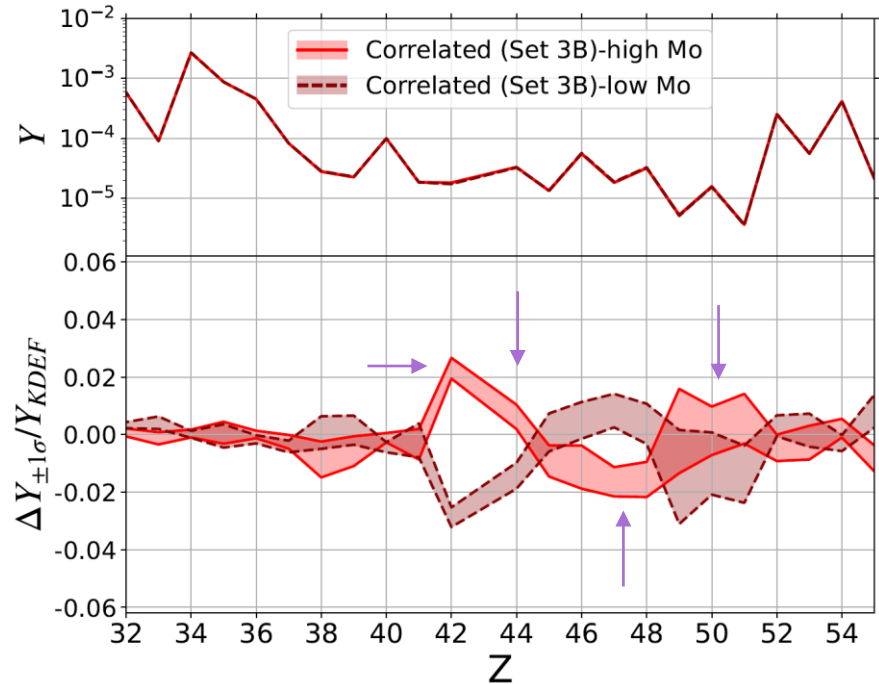


Kedia, et al., [arXiv:2602.12428 \(2026\)](https://arxiv.org/abs/2602.12428)

Correlated neutron capture rates



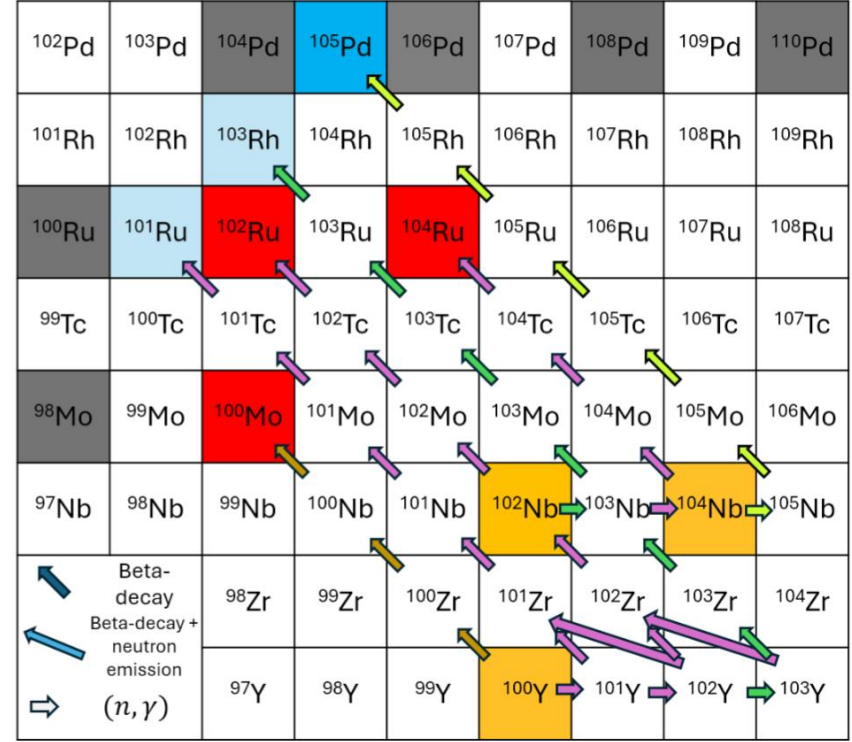
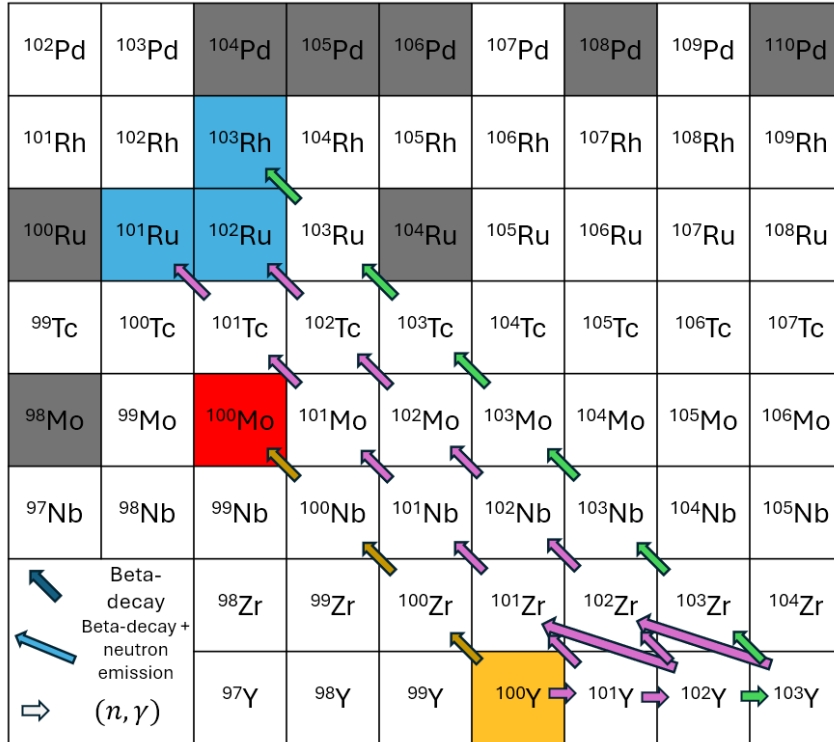
Uncorrelated: Simulations with high Mo (42) do not exhibit an increase or decrease in most other elements.



Correlated: Simulations with high Mo (42) also have high Ru (44), Sb(51) and low Rh (45), Pd (46), Ag (47), Cd (48).

Kedia, et al., [arXiv:2602.12428 \(2026\)](https://arxiv.org/abs/2602.12428)

Correlations in neutron capture rates – Toy model



Kedia, et al., [arXiv:2602.12428 \(2026\)](https://arxiv.org/abs/2602.12428)

Supernova 1987a

First Multi-Messenger event

Distance ~ 50 kpc

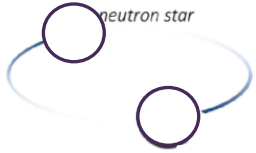
Electromagnetic (EM) and
neutrino emission detected



Bayesian inference of BNS ejecta

Binary properties

$$\vec{x} = \{M_{\text{BIL}}, M_{\text{NS}}, \chi_{\text{BIL}}, \Lambda_{\text{NS}}, \dots\}$$

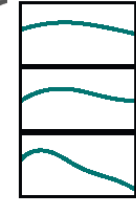
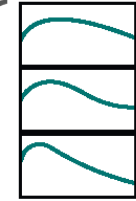


Outflow properties

$$\vec{y} = \{M_{\text{dyn}}, M_{\text{wind}}, v_{\text{dyn}}, v_{\text{wind}}, \dots\}$$



Light curves



⋮

⋮

Fit formulae to
numerical simulations

$$\vec{x}_1$$

BNS merger modelling
(Numerical relativity)

Radiative transfer
simulations

$$\vec{y}_{11}$$

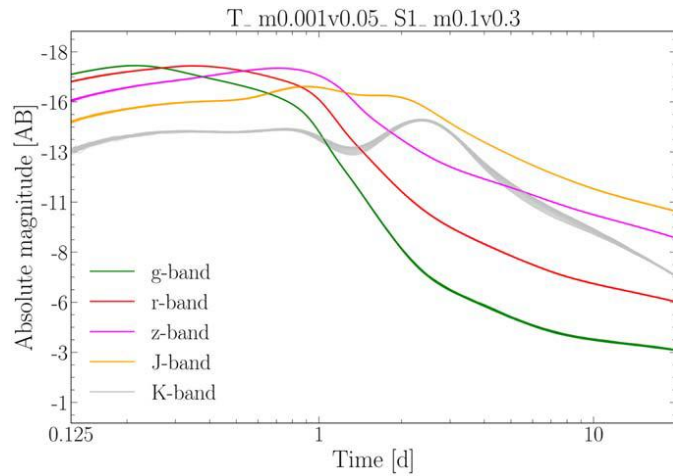
$$\vec{y}_{12}$$

$$\vec{y}_{13}$$

⋮

Ejecta material that
undergoes
nucleosynthesis

Kilonova
modelling



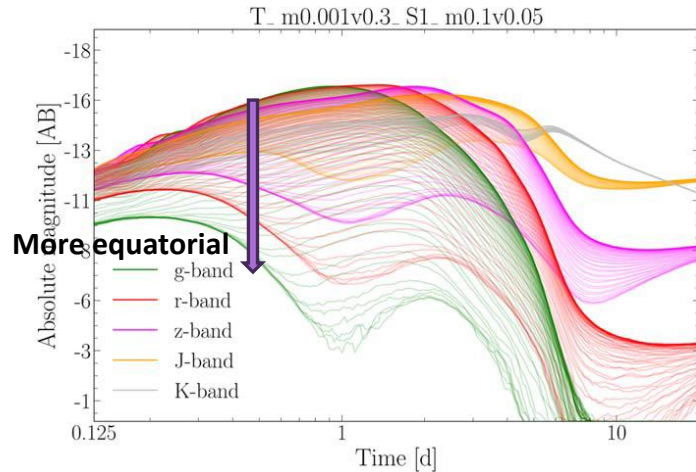
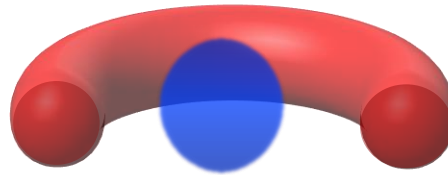
Lanthanide curtaining

A smoking gun for r-process

Curtaining of low wavelength (“blue”) light by Lanthanides (present in the “red” ejecta) produced by the nucleosynthesis.

Top panel: Ejecta more spherically symmetric and less obstructed by the “red” component. Therefore, there is virtually no angular dependence.

Bottom: “Red” component travels further and obstructs light emitted along the equator. Therefore, Lanthanide curtaining is observed. Low wavelength light is suppressed substantially along the equator. High wavelength light passes unaffected (spherically symmetric).

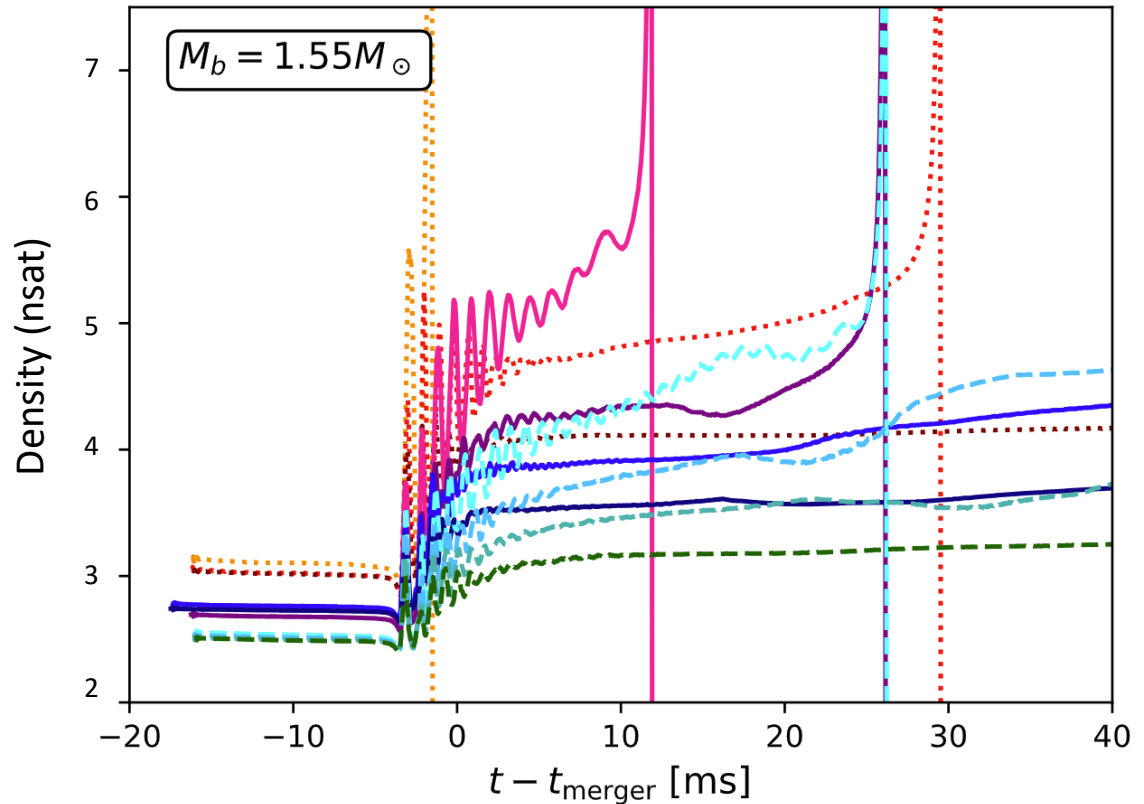


Wollaeger et al ApJ 2021; See also Korobkin+2021, Kasen+2015

Collapse of hypermassive NSs

Postmerger hypermassive
Neutron star remnant
dependence on the
nuclear Equation of State
(i.e. what defines the
neutron star structure)

Vertical drop represents
black hole formation.



Wei Sun, Mathews, Kedia, et al., (in prep)